ice cubes at 10 p.m., *given the one observation we are taking as unassailable, as fully trustworthy, that right now, at 10:30 p.m., you see partially melted cubes.*

This is the pivotal point, so let me explain. The main lesson of the second law of thermodynamics is that physical systems have an overwhelming tendency to be in high-entropy configurations because there are so many ways such states can be realized. And once in such high-entropy states, physical systems have an overwhelming tendency to stay in them. High entropy is the natural state of being. You should never be surprised by or feel the need to explain why any physical system is in a high-entropy state. Such states are the norm. On the contrary, what does need explaining is why any given physical system is in a state of order, a state of low entropy. These states are not the norm. They can certainly happen. But from the viewpoint of entropy, such ordered states are rare aberrations that cry out for explanation. So the one fact in the episode we are taking as unquestionably true—your observation at 10:30 p.m. of low-entropy partially formed ice cubes—is a fact in need of an explanation.

And from the point of view of probability, it is absurd to explain this low-entropy state by invoking the even *lower*-entropy state, the *even less likely* state, that at 10 p.m. there were *even more ordered*, *more fully formed* ice cubes being observed in a *more* pristine, *more* ordered environment. Instead, it is enormously more likely that things began in an unsurprising, totally normal, high-entropy state: a glass of uniform liquid water with absolutely no ice. Then, through an unlikely but every-so-often-expectable statistical fluctuation, the glass of water went against the grain of the second law and evolved to a state of lower entropy in which partially formed ice cubes appeared. This evolution, although requiring rare and unfamiliar processes, completely avoids the even lower-entropy, the even less likely, the even more rare state of having *fully* formed ice cubes. At every moment between 10 p.m. and 10:30 p.m., this strange-sounding evolution has *higher* entropy than the normal ice-melting scenario, as you can see in Figure 6.3, and so it realizes the accepted observation at 10:30 p.m. in a way that is *more likely*—hugely more likely—than the scenario in which fully formed ice cubes melt.[13] That is the crux of the matter.[13]